Question	Answer	Marks	Partial Marks
5(d)(ii)	Both intersections of *their* **(b)** and *their* **(d)(i)**	**2**	**Strict FT** intersection of *their* **(b)** and *their* **(d)(i)** **B1FT** for one correct OR **B2** for 0.27 to 0.28 **and** 2.38 to 2.39
5(e)	Substitutes $x = \sqrt{2}$ into $\frac{1}{2x} - \frac{x}{4}$ OR Identifies $y = 0$ oe OR Correctly manipulates to a single fraction e.g. $\frac{2-x^2}{4x}$ oe seen	**M1**	
	Concludes 'read the graph at $y = 0$' oe OR Manipulates $0 = \frac{1}{2x} - \frac{x}{4}$ oe leading to $x^2 = 2$ OR States $\frac{2-x^2}{4x}$ oe $= 0$ leading to $x^2 = 2$	**A1**	
6(a)	$x^2 + 4x - 21$ final answer	**2**	**B1** for three of $x^2, +7x, -3x, -21$
6(b)(i)	$5q^2(3p^2 - 5q)$ final answer	**2**	**B1** for $5(3p^2q^2 - 5q^3)$ or $q^2(15p^2 - 25q)$ or $q(15p^2q - 25q^2)$ or $5q(3p^2q - 5q^2)$ or for correct answer seen
6(b)(ii)	$(2g + 5k)(2f + 3h)$ final answer	**2**	**B1** for $2g(2f + 3h) + 5k(2f + 3h)$ or $2f(2g + 5k) + 3h(2g + 5k)$ or for correct answer seen
6(b)(iii)	$(9k + m)(9k - m)$ final answer	**2**	**M1** for $(9 + m)(9 - m)$ or for correct answer seen

Question	Answer	Marks	Partial Marks
6(c)	5.5	**4**	**M1** for $5\times3(x-4)+x+2=5\times6$ **M1** for $15x-60+x+2=30$ **FT** *their* first step or $3x-12+\frac{x+2}{5}=6$ If M0M0, **SC1** for $3x - 12 + x + 2 = 30$ oe **M1dep** for $16x=88$ **FT** *their* previous steps
7(a)	$180-\frac{360}{5}$ or $\frac{(5-2)\times180}{5}$ or $\frac{(2\times5-4)\times90}{5}$ or $\frac{5\times180-360}{5}$	**M2**	or **M1** for $\frac{360}{5}$ or $(5-2)\times180$ or $90(2 \times 5 - 4)$ or $3 \times 180 \div 5$ or $6 \times 90 \div 5$ or $5 \times 180 - 360$ If 0 scored, **SC1** for $\frac{5-2\times180}{5}$
7(b)(i)	7.05 or 7.053…	**3**	**M2** for $12 \times \cos54$ oe or **M1** for implicit form or **B1** for length of edge of pentagon = 14.1 to 14.11 If 0 scored, **SC1** for right angle at *M*
7(b)(ii)(a)	22.8 or 22.81 to 22.83… nfww	**3**	**M2** for $\frac{\textit{their}\,\textbf{(b)(i)}}{\cos 72}$ oe or **M1** for implicit form oe or **B1** for *AX* = 36.9 or 36.93 to 36.94
7(b)(ii)(b)	179 or 179.1 to 179.3…	**3**	**M2** for $\frac{1}{2}\times12\times\textit{their AX}\times\sin54$ oe or $\frac{1}{2}\times12\times\textit{their OX}\times\sin108$ oe or $\frac{1}{2}\times\textit{their AX}\times\textit{their OX}\times\sin18$ or $\frac{1}{2}\times12^2\times\sin72+\text{area}\,OBX$ oe or $\frac{1}{2}\times12^2\times\sin72+\text{area}\,OMB+\text{area}\,MBX$ oe or **M1** for a correct method to find area of one relevant triangle *AOB*, *OMB*, *MBX*, *OBX* or *ONX* **seen**
8(a)(i)	15.7 or 15.70...	**4**	**M2** for $16.5^2+12.4^2-2\times16.5\times12.4\times\cos64$ or **M1** for implicit form **A1** for 246 to 247